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OPERSTING SYSTEMS ADVANCED WEEK 12

1. Hands-On Virtualization Lab Report

1. Introduction

Virtualization is a technology that allows one physical computer to run multiple operating systems using virtual machines. A Type 2 hypervisor is software installed on an existing operating system that allows users to create and manage virtual machines.

The aim of this practical is to install Oracle VirtualBox, create a virtual machine, install Ubuntu Server Linux, configure its hardware resources, and test the snapshot and restore features.

2. Requirements

The following software and hardware are required:

- A computer running Windows.
- Oracle VirtualBox software.
- Ubuntu Server ISO image.
- At least 8 GB of RAM recommended.
- Enough free disk space for the virtual machine.

3. Procedure

Step 1: Install VirtualBox

I visited the official VirtualBox website at https://www.virtualbox.org/ and downloaded the software for Windows. I ran the installer and followed the instructions to complete the installation.

Step 2: Download Ubuntu Server

I visited https://ubuntu.com/download/server and downloaded the Ubuntu Server ISO image. This file was used to install Linux inside the virtual machine.

Step 3: Create a Virtual Machine

I opened VirtualBox and clicked **New** to create a virtual machine. I named it Ubuntu Server VM and selected Linux as the operating system type. I then selected the downloaded Ubuntu Server ISO image.

Step 4: Configure the Virtual Machine

I configured the virtual machine using the following settings:

Component	Configuration
Virtual CPU cores	2 cores
Memory (RAM)	2 GB
Virtual hard disk	25 GB
Disk type	Dynamically allocated
Operating system	Ubuntu Server Linux
Network	NAT

The virtual CPU cores provide processing power, while the allocated RAM allows Ubuntu Server to run. The virtual hard disk provides storage for the operating system and files. NAT allows the virtual machine to access the internet through the host computer's network connection.

Step 5: Install Ubuntu Server

I started the virtual machine and followed the Ubuntu Server installation instructions. I selected the language and keyboard layout, configured the network, selected the virtual disk, and created a username and password.

After completing the installation, I restarted the virtual machine and logged in to Ubuntu Server.

Step 6: Take a Snapshot

After installing Ubuntu Server, I shut down the virtual machine and opened the Snapshots section in VirtualBox. I created a snapshot named **Clean Ubuntu Installation**.

The snapshot saved the virtual machine's state before any additional changes were made. It would allow me to return to this state if something went wrong.

Step 7: Make a Change to the Virtual Machine

I started Ubuntu Server and logged in. I used the following commands to create a test file:

```
echo "Virtualization Lab Test" > labtest.txt
cat labtest.txt
```

The output displayed was:

```
Virtualization Lab Test
```

This confirmed that the file had been created successfully.

Step 8: Restore the Snapshot

I shut down the virtual machine and opened the Snapshots section in VirtualBox. I selected the snapshot named **Clean Ubuntu Installation**, clicked Restore, and confirmed the action.

After starting Ubuntu Server again, I entered the following command:

```
ls labtest.txt
```

The file was not found because it had been created after the snapshot was taken. This demonstrated that restoring the snapshot returned the virtual machine to its previous state.

4. Results and Observations

The practical demonstrated that VirtualBox can run Ubuntu Server inside a virtual machine without replacing the host operating system. The virtual machine was configured with two CPU cores, 2 GB of RAM, and a 25 GB virtual disk.

The snapshot feature was also tested successfully in the procedure. Creating the test file and then restoring the snapshot demonstrated how snapshots can help recover from unwanted changes.

5. Conclusion

In conclusion, this practical helped me understand how virtualization works and how to use a Type 2 hypervisor. I learned how to create a virtual machine, allocate hardware resources, install Ubuntu Server, create a snapshot, and restore the virtual machine to an earlier state.

Virtualization is useful because it allows users to test operating systems and software in an isolated environment without directly affecting the main computer. Snapshots also make it easier to recover from mistakes during testing and system administration.

6. Evidence

The following screenshots should be attached to demonstrate the practical work:

1. VirtualBox installed on the computer.
2. Virtual machine hardware configuration.
3. Ubuntu Server running successfully.
4. The snapshot named *Clean Ubuntu Installation*.
5. The test file displayed before restoring the snapshot.
6. The terminal output showing that the test file is missing after restoration.

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